

JOINT DISTRIBUTION-INFORMED SHAPLEY VALUES FOR SPARSE COUNTERFACTUAL EXPLANATIONS

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ABSTRACT

Counterfactual explanations (CE) are a broadly used way to describe how input changes can move a model prediction toward a desired outcome, but across CE generators they often alter more features than the task requires, limiting clarity and practical use. We introduce *COLA*, a post-hoc framework that is agnostic to the predictive model class and to the CE generator: given factual and counterfactual sets from an existing CE method, *COLA* computes an optimal-transport (OT) coupling and uses it to drive a joint-probability-informed Shapley attribution (*p*-*SHAP*) for selecting a compact set of edits that preserves the target effect. Theoretically, OT minimizes an upper bound on the W_1 divergence between factual and counterfactual outcomes, and under mild conditions the refined counterfactuals do not move farther from the factuals than the aligned originals. Empirically, on four tabular datasets spanning twelve classifiers and five CE generators, *COLA* achieves the same target effects with only 26–45% of the original feature edits. On tractable small-scale benchmarks, *COLA* is near-optimal.

The experiment code and software will be released upon acceptance.

1 BACKGROUND

Explainable Artificial Intelligence (XAI) supports transparency and trust across artificial intelligence systems deployed in diverse decision settings (Arrieta et al., 2020; Das & Rad, 2020). Within this area, feature attributions (FA) methods, including Shapley values (Sundararajan & Najmi, 2020; Lundberg & Lee, 2017), quantify how input features contribute to a machine learning (ML) model’s output, thereby making complex predictive behavior easier to inspect. In healthcare, for example, Shapley values can highlight factors such as age and medical history, helping clinicians interpret model decisions (Ter-Minassian et al., 2023; Nohara et al., 2022); analogous attribution questions arise in many other high-stakes and consumer-facing applications. A complementary line of work, counterfactual explanations (CE) (Wachter et al., 2017; Guidotti, 2022), explains decisions by describing input changes that would lead to different outcomes. Hundreds of CE algorithms have been proposed (Guidotti, 2022; Verma et al., 2020), and this breadth reflects the fact that no single generator is appropriate for all models, tasks, data organizations, and user goals. Some methods search for one counterfactual per factual instance, others operate on groups of instances, and still others generate global directions for an entire dataset (Rawal & Lakkaraju, 2020; Ley et al., 2022; 2023; Carrizosa et al., 2024). Distributional approaches instead treat factual instances as an empirical distribution and seek counterfactual distributions with comparable shape and cost (You et al., 2025). The modeling assumptions are similarly varied: some generators require differentiability, while others are tailored to trees, ensembles, or black-box access.

Problem Description and Challenges We study a broad refinement problem for factual data $\mathbf{x}$ that sits at the intersection of FA and CE, building on the extensive foundations established in the literature (see Appendix A and Appendix A). Both tools are central to interpretable and accountable AI, but each is incomplete on its own: FA can identify influential variables without offering actionable changes, whereas CE can recommend concrete changes that include unnecessary or impractical edits. The question we address is therefore not tied to a single dataset, model family, or CE generator; rather, it asks how an existing counterfactual proposal $\mathbf{r}$ can be converted into a smaller action plan while retaining its intended outcome.

Given a (group of) factual instance(s), how can we devise an action plan that requires the least feature modifications to achieve a desired counterfactual outcome?

$\mathbf{x}$	$\mathbf{z}'$	$\mathbf{z}''$	$\mathbf{y}^*$
$\textcircled{\$}$ $\blacktriangleleft$ $\textcircled{\text{R}}$	$\textcircled{\$}$ $\blacktriangleleft$ $\textcircled{\text{R}}$	$\textcircled{\$}$ $\blacktriangleleft$ $\textcircled{\text{R}}$	$\textcircled{\text{R}}$
200 5 No	250 8 Yes	200 5 No	Yes
150 3 No	150 3 No	150 7 Yes	Yes
100 2 No	350 9 Yes	100 2 No	Yes
150 6 No	150 6 No	350 6 Yes	Yes

Figure 1: [Example: User engagement on an e-commerce platform] A platform aims to increase user registrations. The platform has collected data on user interactions, such as the amount of money spent ($\textcircled{\$}$), the number of clicks ($\blacktriangleleft$), and whether the user has registered ($\textcircled{\text{R}}$). In the original data ($\mathbf{x}$), no users are registered. Action plans $\mathbf{z}'$ and $\mathbf{z}''$ adjust user characteristics to achieve the desired outcome ($\mathbf{y}^*$) of full registration. Both plans achieve a half counterfactual effect, but $\mathbf{z}''$ requires fewer modifications compared to $\mathbf{z}'$.

Three challenges make this refinement problem broadly relevant. First, because CE objectives differ across instance-level, group-level, global, and distributional settings, a refinement method should not depend on a single generator being universally appropriate. Second, the method should avoid strong structural assumptions on the predictive model, such as differentiability or a particular architecture, so that it can be applied to a wide range of classifiers. Third, ordinary FA scores can be misleading for action selection when they are disconnected from the counterfactual proposal. In other words, it is not effective to perform FA independently of CE to select the most important features to change. We will demonstrate later that this decoupling can produce counterproductive feature modifications (also referred to as actions from users/stakeholders) in Result II of Section 6, because globally important features need not coincide with the feature changes that move the model toward the desired counterfactual outcome.

Main Contributions We introduce COUNTERFACTUALS WITH LIMITED ACTIONS (COLA), a modular post-hoc framework for refining CE outputs across model classes and CE generators by using an optimal transport (OT)–induced coupling between factual and counterfactual sets to guide Shapley attributions. This coupling leads to p -SHAP, which recovers several commonly used Shapley variants under appropriate choices of joint distribution and provides a common interface for attribution-driven edit selection. We prove that OT minimizes an upper bound on the W_1 divergence, and that under mild conditions the refined counterfactuals are no farther from the factuals than the aligned counterfactual proposals. Empirically, across four tabular datasets, twelve models, and five CE methods, COLA with p -SHAP achieves the same target effects with only 26–45% of the original feature edits. In tractable small-scale benchmarks, COLA is near-optimal.

To our best knowledge, this is the first systematic treatment of the refinement problem in Figure 1 that is designed to operate *without specifying certain CE algorithms and models*.

2 PROBLEM FORMULATION

We now formulate the refinement problem illustrated in Figure 1 in a way that covers single-instance, group, and distributional counterfactual settings. Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ denote a black-box ML model, and let $\mathbf{x}$ be observed factual data with n rows and d columns ($\mathbf{x} \in \mathbb{R}^{n \times d}$, $n \geq 1$, and $d \geq 1$). Let $\mathbf{y}^*$ be the target model output ($\mathbf{y}^* \in \mathbb{R}^m$, $m \geq 1$). The optimization seeks counterfactual data $\mathbf{z}$ ($\mathbf{z} \in \mathbb{R}^{n \times d}$) that use at most C feature changes while making the model output as close as possible to $\mathbf{y}^*$. Let D denote a divergence function measuring dissimilarity between two entities. The problem

is:

$$\min_{\mathbf{c}, \mathbf{z}} D(f(\mathbf{z}), \mathbf{y}^*), \quad (1a)$$

$$\text{s.t. } D(\mathbf{z}, \mathbf{x}) \leq \epsilon, \quad (1b)$$

$$\sum_{i=1}^n \sum_{k=1}^d c_{ik} \leq C, \quad (1c)$$

$$z_{ik} \leq x_{ik}(1 - c_{ik}) + M c_{ik} \quad i = 1, \dots, n, \quad k = 1, \dots, d, \quad (1d)$$

$$z_{ik} \geq x_{ik}(1 - c_{ik}) - M c_{ik} \quad i = 1, \dots, n, \quad k = 1, \dots, d. \quad (1e)$$

The objective equation (1a) and constraint equation (1b) capture the usual CE tradeoff in equation (1b): $\mathbf{z}$ should make $f(\mathbf{z})$ close to $\mathbf{y}^*$ while remaining close to $\mathbf{x}$. Such a $\mathbf{z}$ can then serve as a contrastive reference for why $f(\mathbf{x})$ does not achieve $\mathbf{y}^*$, relative to a CE proposal $\mathbf{r}$ for the same $\mathbf{x}$. To keep the formulation applicable across evaluation choices, we do not restrict D to a single divergence; examples include Euclidean distance, OT, maximum mean discrepancy (MMD), and differences in the mean or median of model outcomes. Constraints equation (1c)–equation (1e) add an action budget to this CE objective. In addition to optimizing $\mathbf{z}$, we optimize an indicator matrix $\mathbf{c}$, where z_{ik} is fixed to x_{ik} iff $c_{ik} = 0$. Constraint equation (1c) limits the total number of changed entries to C . From equation (1d) and equation (1e), $c_{ik} = 0$ forces $z_{ik} = x_{ik}$, whereas $c_{ik} = 1$ allows z_{ik} to vary within the sufficiently large bound controlled by M .

To make equation (1) tractable across CE generators and model classes, we use FA to identify entries of $\mathbf{x}$ that should receive the limited modifications encoded by $\mathbf{c}$. The next section reviews Shapley value methods for FA; these methods, together with our proposed variant, form the attribution component of COLA for constructing the refined counterfactual $\mathbf{z}$. The problem is computationally difficult even when $d = 1$ for linear models, see Appendix B.

3 PRELIMINARIES ON SHAPLEY VALUE

This section reviews Shapley value methods that can serve as attribution modules within COLA, including both standard variants and the joint-probability-informed variant introduced later. We begin with the game-theoretic Shapley value and then describe commonly used instantiations for FA. A broader survey of Shapley value methods appears in (Sundararajan & Najmi, 2020).

In cooperative game theory, a coalitional game is defined by a finite set of players–features in our setting—denoted by $\mathcal{F} = \{1, 2, \dots, d\}$, together with a characteristic function v . The function $v : 2^{\mathcal{F}} \rightarrow \mathbb{R}$ assigns each subset $S \subseteq \mathcal{F}$ a payoff $v(S)$, with $v(\emptyset) = 0$. The Shapley value uses this set function to distribute the grand-coalition payoff $v(\mathcal{F})$ across individual players in an axiomatic way, which makes it a natural tool for feature-level attribution across many model types.

Before defining the Shapley value, let $\Delta(k, S)$ denote the marginal contribution of player (feature) k to coalition S . This quantity measures the additional payoff obtained when feature k is added to the coalition:

$$\Delta(k, S) = v(S \cup k) - v(S). \quad (2)$$

The Shapley value ϕ_i for a player i is then defined in equation (3).

$$\phi_k = \frac{1}{d} \sum_{S \subseteq \mathcal{F} \setminus \{k\}} \binom{d-1}{|S|}^{-1} \Delta(k, S). \quad (3)$$

The value ϕ_k averages the normalized marginal contributions $\Delta(k, S)$ of feature k over all subsets S that exclude k . The inverse binomial coefficient $\binom{d-1}{|S|}^{-1}$ weights coalitions by size, and the final division by d yields an axiomatic allocation of the total payoff across features. In FA, this allocation represents each feature’s contribution relative to the full collection of inputs $\mathbf{x}$ for a data point such as $\mathbf{x}_i$.

For an arbitrary data point $\mathbf{x}_i$, we write $\mathbf{x}_{i,S}$ for the components indexed by S and $\mathbf{x}_{i,\mathcal{F} \setminus S}$ for the remaining components; the same convention applies to other vectors, including counterfactual references $\mathbf{r}$ and refined outputs $\mathbf{z}$. We next review Shapley variants that differ primarily in how they represent absent features and contrastive references $\mathbf{r}$.

Baseline Shapley (B-SHAP). This approach uses a specific baseline data point $\mathbf{r}_j$ as the reference for any $\mathbf{x}_i$. The set function is

$$v_B^{(i)}(S) = f(\mathbf{x}_{i,S}; \mathbf{r}_{j, \mathcal{F} \setminus S}) - f(\mathbf{r}_j), \quad (4)$$

where an absent feature is replaced by its value in the reference baseline $\mathbf{r}$ (Lundberg & Lee, 2017; Sun & Sundararajan, 2011; Merrick & Taly, 2020). This construction is appropriate when an exact alignment between $\mathbf{x}$ and $\mathbf{r}$ is available or assumed.

Random Baseline Shapley (RB-SHAP). This variant replaces a single reference point with a background distribution, as in (Lundberg & Lee, 2017) and the widely used ‘‘SHAP’’ library. The corresponding set function is

$$v_{RB}^{(i)}(S) = \mathbb{E}_{\mathbf{x}' \sim \mathcal{D}} [f(\mathbf{x}_{i,S}; \mathbf{x}'_{\mathcal{F} \setminus S})] - \mathbb{E}_{\mathbf{x}' \sim \mathcal{D}} [f(\mathbf{x}')]. \quad (5)$$

where $\mathcal{D}$ is a background distribution, commonly instantiated as the training dataset (Lundberg & Lee, 2017; Merrick & Taly, 2020).

Counterfactual Shapley (CF-SHAP). This extension of RB-SHAP takes $\mathcal{D}$ to be a counterfactual distribution associated with each factual instance $\mathbf{x}_i$ ($i = 1, 2, \dots, n$).

$$v_{CF}^{(i)}(S) = \mathbb{E}_{\mathbf{r} \sim \mathcal{D}(\mathbf{x}_i)} [f(\mathbf{x}_{i,S}; \mathbf{r}_{\mathcal{F} \setminus S})] - \mathbb{E}_{\mathbf{r} \sim \mathcal{D}(\mathbf{x}_i)} [f(\mathbf{r})]. \quad (6)$$

This approach has been adopted in (Albini et al., 2022; Kommiya Mothilal et al., 2021) and has shown advantages for contrastive FA. Its scope depends on having a probabilistic alignment between $\mathbf{x}$ and $\mathbf{r}$, whereas our refinement setting must also construct $\mathbf{z}$ when that alignment is not directly supplied.

4 THE PROPOSED JOINT-PROBABILITY-INFORMED SHAPLEY (p -SHAP) AND ITS THEORETICAL ASPECTS

(Proposed) p -SHAP We generalize the constructions in equation (4)–equation (6) by introducing an algorithm A_{Prob} that returns a joint probability over factual and counterfactual instances.¹ This definition makes the alignment mechanism explicit while leaving its source modular:

$$v^{(i)}(S) = \mathbb{E}_{\mathbf{r} \sim p(\mathbf{r}|\mathbf{x}_i)} [f(\mathbf{x}_{i,S}; \mathbf{r}_{\mathcal{F} \setminus S})] - \mathbb{E}_{\mathbf{r} \sim p(\mathbf{r})} [f(\mathbf{r})], \quad (7a)$$

$$\text{s.t. } p = A_{\text{Prob}}(\mathbf{x}, \mathbf{r}). \quad (7b)$$

Crucially, unlike methods that rely only on random baselines, p -SHAP can address the alignment problem by identifying an optimal coupling via OT. This views feature attribution not only as a prediction decomposition, but also as a transport-informed attribution problem that favors low-cost factual-to-counterfactual explanations.

Remark that A_{Prob} is independent of the CE algorithm, but only depends on the generated counterfactual $\mathbf{r}$. This separation lets p -SHAP use counterfactual information without inheriting the internal assumptions of the CE generator, which distinguishes it from CF-SHAP in settings where the generation mechanism differs across tasks or model classes. Contrary to common expectations, we demonstrate that OT can be more effective than relying on a counterfactual distribution defined by a CE generation mechanism as done by Albini et al. (2022), in Result II of Section 6 later.

In this paper, we instantiate A_{Prob} with the OT problem, equivalently the entropic form of the 2-Wasserstein divergence, defined below. The resulting transportation plan $\mathbf{p}_{\text{OT}}$ serves as the joint distribution between $\mathbf{x}$ and $\mathbf{r}$ in p -SHAP.

$$\mathbf{p}_{\text{OT}} = \arg \min_{\mathbf{p} \in \Pi(\boldsymbol{\mu}, \boldsymbol{\nu})} \sum_{i=1}^n \sum_{j=1}^m p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2 + \varepsilon \sum_{i=1}^n \sum_{j=1}^m p_{ij} \log \left(\frac{p_{ij}}{\mu_i \nu_j} \right). \quad (8)$$

Here $\boldsymbol{\mu}$ and $\boldsymbol{\nu}$ denote the marginal distributions of $\mathbf{x}$ and $\mathbf{r}$, respectively, and $\Pi(\boldsymbol{\mu}, \boldsymbol{\nu})$ is the set of admissible joint distributions, or transport plans. The term $\varepsilon \sum_{i=1}^n \sum_{j=1}^m p_{ij} \log(p_{ij}/(\mu_i \nu_j))$ is

¹First, p -SHAP degrades to B-SHAP in equation (4) when A_{Prob} defines a joint distribution between $\mathbf{x}$ and $\mathbf{r}$ that indicates an $i \leftrightarrow j$ alignment of for any $\mathbf{x}_i, \mathbf{r}_j$. Second, p -SHAP degrades to RB-SHAP in equation (5) when A_{Prob} is defined to be independent of CE but associates with an arbitrary distribution $\mathcal{D}$. Third, p -SHAP degrades to CF-SHAP in equation (6) when A_{Prob} is built upon a known distribution of CE.

entropic regularization, with coefficient $\varepsilon \geq 0$. When $\varepsilon > 0$, this regularization supports faster computation of OT, which is useful for applying the same framework across multiple datasets and models.

Theoretical Aspects of p -SHAP OT minimizes the total feature modification cost for transporting factual instances $\mathbf{x}$ toward counterfactual instances $\mathbf{r}$ under the learned alignment. This is aligned with our goal of finding a limited set of feature modifications that preserves the counterfactual outcome at low cost. We can formalize this connection under a Lipschitz continuity assumption on the predictive model f . In Theorem 4.1 below (proof in Appendix C), we show that the transportation plan $\mathbf{p}_{\text{OT}}$ used in p -SHAP minimizes an upper bound on the divergence between $f(\mathbf{x})$ and $\mathbf{y}^*$. Specifically, the 1-Wasserstein distance between $f(\mathbf{x})$ and $\mathbf{y}^*$ is bounded by the Lipschitz constant multiplied by the square root of the minimized expected cost of moving $\mathbf{x}$ toward $\mathbf{r}$, i.e. $\sum_{i,j} p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2$, where p_{ij} ($j = 1, 2, \dots, m$) specifies how $\mathbf{x}_i$ is aligned with one or more counterfactual rows $\mathbf{r}_j$. Practically, the OT plan $\mathbf{p}_{\text{OT}}$ gives p -SHAP a transport-informed strategy for selecting edits: among plans in Π , it yields the smallest expected modification cost and therefore the tightest cost-based upper bound on the counterfactual-effect violation $W_1(f(\mathbf{x}), \mathbf{y}^*)$.

Theorem 4.1 (p -SHAP Towards Counterfactual Effect). *Consider the 1-Wasserstein divergence W_1 , i.e. $W_1(f(\mathbf{x}), \mathbf{y}^*) = \min_{\pi \in \Pi} \sum_{i=1}^n \sum_{j=1}^m \pi_{ij} |f(\mathbf{x}_i) - \mathbf{y}_j^*|$. Suppose the counterfactual outcome $\mathbf{y}^*$ is fully achieved by $\mathbf{r}$, i.e. $\mathbf{y}_j^* = f(\mathbf{r}_j)$ ($j = 1, 2, \dots, m$). Assume that the model $f: \mathbb{R}^d \rightarrow \mathbb{R}$ is Lipschitz continuous with Lipschitz constant L . The expected absolute difference in model outputs between the factual and counterfactual instances, weighted by $\mathbf{p}_{\text{OT}}$ (with $\varepsilon = 0$), is bounded by:*

$$W_1(f(\mathbf{x}), \mathbf{y}^*) \leq L \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{\text{OT}} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2} \leq L \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2}, \quad \forall \mathbf{p} \in \Pi.$$

Namely, $\mathbf{p}_{\text{OT}}$ minimizes the upper bound of $W_1(f(\mathbf{x}), \mathbf{y}^*)$, where the upper bound is based on the expected feature modification cost.

Although Theorem 4.1 bounds transport cost rather than the discrete sparsity objective directly, it provides a convex proxy for the $\mathcal{NP}$ -hard L_0 selection problem. By concentrating attribution mass on efficient factual-to-counterfactual paths, the OT coupling guides the greedy selection in Algorithm 1 toward discarding non-essential features. In addition, p -SHAP has an interventional interpretation for attributing the effect of feature modifications, as stated in Theorem 4.2 below (proof in Appendix D).

Theorem 4.2 (Interventional Effect of p -SHAP). *For any subset $S \subseteq \mathcal{F}$ and any $\mathbf{x}_i$ ($i = 1, 2, \dots, n$), $v^{(i)}(S)$ represents the causal effect of the difference between the expected value of $f(\mathbf{r})$ under the intervention on features S and the unconditional expected value of $f(\mathbf{r})$. Mathematically, this is expressed as:*

$$\mathbb{E}[f(\mathbf{r})] + v^{(i)}(S) = \mathbb{E}[f(\mathbf{r}) | do(\mathbf{r}_S = \mathbf{x}_{i,S})].$$

Furthermore, p -SHAP preserves the desirable Shapley axioms used by B-SHAP and RB-SHAP, making it suitable as a general attribution module. We omit the proof and refer to (Sundararajan & Najmi, 2020; Lundberg & Lee, 2017) for the relevant Shapley axioms.

5 THE ALGORITHMIC FRAMEWORK COLA

Sketch The algorithmic framework COLA, stated in Algorithm 1, is designed to solve equation (1) through four modular algorithm categories. We explain Algorithm 1 below and illustrate the workflow in Figure 2.

Line 1 (*Applying a CE algorithm to find a counterfactual $\mathbf{r}$*). The CE algorithm A_{CE} takes the model f , the factual data $\mathbf{x}$, the target outcome $\mathbf{y}^*$, and the tolerance ϵ as input. It returns a counterfactual set $\mathbf{r}$ that remains close to $\mathbf{x}$ while satisfying $\mathbf{y}^* = f(\mathbf{r})$ and $D(\mathbf{r}, \mathbf{x}) \leq \epsilon$.

Line 2 (*Seeking a joint distribution of $\mathbf{x}$ and $\mathbf{r}$*). The algorithm A_{Prob} takes $\mathbf{x}$ and $\mathbf{r}$ as input and outputs a matrix representing a joint distribution over the n factual rows and m counterfactual rows. This matrix $\mathbf{p}$ encodes an alignment, or matching, between factual and counterfactual data and is later

Algorithm 1 COUNTERFACTUALS WITH LIMITED ACTIONS (COLA)

Require: Model f , factual $\mathbf{x} \in \mathbb{R}^{n \times d}$, target $\mathbf{y}^* \in \mathbb{R}^m$, ϵ , and C
Ensure: Action plan $\mathbf{c} \in \mathbb{R}^{n \times d}$ and correspondingly a refined counterfactual $\mathbf{z} \in \mathbb{R}^{n \times d}$

- 1: Use $A_{\text{CE}}(f, \mathbf{x}, \mathbf{y}^*, \epsilon)$ to obtain $\mathbf{r} \in \mathbb{R}^{m \times d}$, with $\mathbf{y}^* = f(\mathbf{r})$ and $D(\mathbf{r}, \mathbf{x}) \leq \epsilon$.
- 2: Use $A_{\text{Prob}}(\mathbf{x}, \mathbf{r})$ to obtain the joint distribution matrix $\mathbf{p} \in \mathbb{R}_+^{n \times m}$.
- 3: Use $A_{\text{Shap}}(\mathbf{x}, \mathbf{r}, \mathbf{p})$ to obtain the shapley values $\phi \in \mathbb{R}^{n \times d}$ for each element of $\mathbf{x}$.
- 4: Normalize the element-wise absolute values of ϕ , i.e., $\varphi_{ik} \leftarrow |\phi_{ik}| / \|\phi\|_1$ ($\varphi \in \mathbb{R}_+^{n \times d}$).
- 5: Use $A_{\text{Value}}(\mathbf{r}, \mathbf{p})$ to obtain matrix $\mathbf{q} \in \mathbb{R}^{n \times d}$.
- 6: For $\mathbf{c} \in \{0, 1\}^{n \times d}$, $c_{ik} \leftarrow 0$ ($i = 1, \dots, n$, $k = 1, \dots, d$).
- 7: Sample C pairs (i, k) according to the probability matrix φ , and let $c_{ik} = 1$ for them.
- 8: Let $\mathbf{z} \leftarrow \mathbf{x}$ ($\mathbf{z} \in \mathbb{R}^{n \times d}$).
- 9: **for** $i \leftarrow 1$ to n **do**
- 10: **for** $k \leftarrow 1$ to d **do**
- 11: **if** $c_{ik} = 1$ **then**
- 12: $z_{ik} \leftarrow q_{ik}$
- 13: **end if**
- 14: **end for**
- 15: **end for**
- 16: **return** $\mathbf{c}$ and $\mathbf{z}$

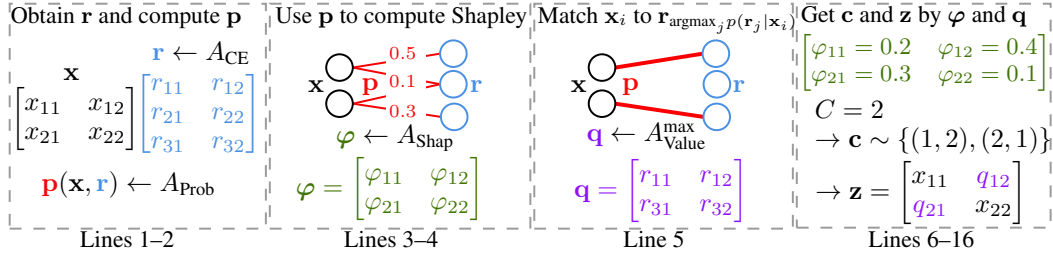


Figure 2: [An illustration of COLA] This figure illustrates how COLA constructs the action indicator $\mathbf{c}$ and refined counterfactual $\mathbf{z}$ for equation (1). For clarity, line 5 is shown with $A_{\text{Value}}^{\max}$. In lines 6–16, $C = 2$, so the sampling step selects exactly two modification positions according to the probability matrix φ .

used in Line 5 to compute candidate values for $\mathbf{z}$. As discussed in Section 3, when the row alignment is unknown, A_{Prob} can be instantiated by OT to obtain a joint distribution with minimal transport distance between $\mathbf{x}$ and $\mathbf{r}$. When reliable alignment information is available from the application or generator, the same interface can instead use a joint distribution that reflects that alignment.

Lines 3–4 (p -SHAP FA). We apply equation (7) to compute Shapley values for $\mathbf{x}$. The joint distribution $\mathbf{p}$ can be used to align each factual row with one or more counterfactual rows, making the selected rows in $\mathbf{r}$ representative contrastive references for the corresponding row in $\mathbf{x}$. This alignment is optional in the interface but, as the experiments show, materially affects contrastive FA. We then take element-wise absolute values of the Shapley matrix for $\mathbf{x}$ and normalize them so that all entries sum to one. The resulting matrix φ is the attribution-based policy used for action selection.

Line 5 (Computing feature values). The algorithm A_{Value} takes the counterfactual set $\mathbf{r}$ and joint distribution $\mathbf{p}$ as input. For each row i in $\mathbf{x}$, A_{Value} selects one or more rows in $\mathbf{r}$ as references for value replacement. It returns a matrix $\mathbf{q} \in \mathbb{R}^{n \times d}$, where each q_{ik} is a candidate counterfactual value for x_{ik} ($\forall i, k$). We consider $A_{\text{Value}}^{\max}$ for single-row selection and $A_{\text{Value}}^{\text{avg}}$ for multi-row averaging.

For any row $\mathbf{x}_i$, $A_{\text{Value}}^{\max}$ selects the row of $\mathbf{r}$ with the highest alignment probability.

$$\mathbf{q} = A_{\text{Value}}^{\max}(\mathbf{r}, \mathbf{p}), \text{ where } q_{ik} = r_{\tau(i),k} \text{ and } \tau(i) = \arg \max_{j=1,2,\dots,m} p_{ij}. \quad (9)$$

The algorithm A^{avg} computes q_{ik} as a convex combination, i.e., a weighted average of $r_{1k}, r_{2k}, \dots, r_{mk}$.

$$\mathbf{q} = A_{\text{Value}}^{\text{avg}}(\mathbf{r}, \mathbf{p}), \text{ where } q_{ik} = \sum_{j=1}^m \left(\frac{p_{ij}}{\sum_{j'=1}^m p_{ij'}} \right) r_{jk}. \quad (10)$$

Lines 6–16. Because the non-negative matrix φ is normalized to sum to one, it can be treated as a policy over candidate modification positions in $\mathbf{x}$. Line 7 samples entries with $c_{ik} = 1$ according to this policy. For each selected pair (i, k) , x_{ik} is replaced by q_{ik} , and the resulting matrix is returned as $\mathbf{z}$ together with the action indicator $\mathbf{c}$ for equation (1).

By Theorem 4.1, COLA is designed to reduce the dissimilarity between $f(\mathbf{z})$ and $\mathbf{y}^*$ by applying edits selected through feature attributions that are aligned with the counterfactual effect. The theorem below (proof in Appendix E) shows that the refined $\mathbf{z}$ produced by COLA remains no farther from the factual input than the aligned counterfactual proposal under the Frobenius norm. The statement applies for arbitrary $n, m \geq 1$, and the special case $n = m$ recovers the common setting where factual and counterfactual rows can be deterministically paired; empirical support across the evaluated scenarios appears in Table 3.

Theorem 5.1 (Counterfactual Proximity). *Let $\mathbf{q} \in \mathbb{R}^{n \times d}$ be the aligned reference matrix derived from the counterfactual set $\mathbf{r} \in \mathbb{R}^{m \times d}$ via the value assignment step $\mathbf{q} = A_{\text{Value}}(\mathbf{r}, \mathbf{p})$, where $\mathbf{p}$ is the coupling matrix. For any dimensions $n, m \geq 1$, the refined counterfactual $\mathbf{z}$ constructed by the COLA framework satisfies:*

$$\|\mathbf{z} - \mathbf{x}\|_F \leq \|\mathbf{q} - \mathbf{x}\|_F.$$

This inequality guarantees that the refined counterfactual $\mathbf{z}$ is no farther from the factual input $\mathbf{x}$ than the aligned counterfactual proposal $\mathbf{q}$, so the refinement step does not introduce additional divergence.

Corollary 5.2. *In the special case where $n = m$ and the OT plan $\mathbf{p}$ corresponds to a deterministic permutation σ (i.e., obtained without entropic regularization, $\varepsilon = 0$), the matrix $\mathbf{q}$ becomes a row-permuted version of $\mathbf{r}$ (denoted as $\mathbf{r}_\sigma$, where $q_i = r_{\sigma(i)}$). In this setting, the bound simplifies to:*

$$\|\mathbf{z} - \mathbf{x}\|_F \leq \|\mathbf{r}_\sigma - \mathbf{x}\|_F,$$

recovering the intuition that the refined counterfactual is at least as close to the facts as the original counterfactual set after reordering by σ .

Complexity of COLA Let $\mathcal{O}(M_{\text{CE}})$ denote the complexity of the chosen CE generator A_{CE} . For A_{Shap} , we consider weighted linear regression for estimating Shapley values and denote the number of sampled subsets by M_{Shap} . The resulting complexity of COLA, expressed in terms of n, m, d , and the entropic OT regularization parameter ε , is $\mathcal{O}(M_{\text{CE}}) + \mathcal{O}(nm \log(1/\varepsilon)) + \mathcal{O}(ndM_{\text{Shap}}) + N$ where $N = \mathcal{O}(nm) + \mathcal{O}(nd)$ when $A_{\text{Value}}^{\text{max}}$ is used and $N = \mathcal{O}(nmd)$ when $A_{\text{Value}}^{\text{avg}}$ is used. Appendix F gives the detailed derivation.

6 NUMERICAL RESULTS

This section evaluates COLA for the refinement problem in equation (1), setting $\mathbf{y}^* = f(\mathbf{r})$ where $\mathbf{r}$ is produced by a CE method A_{CE} . To test the framework under multiple outcome-comparison views, we use four divergence functions: OT measures distribution-level distance, MMD measures discrepancy between distributions in a high-dimensional feature space, and absolute mean difference (MeanD) and absolute median difference (MedianD) measure differences in the mean and median, respectively. The numerical results are designed to show: I) COLA’s effectiveness for modification minimality. II) p-SHAP’s superior performance than other Shapley methods towards modification minimality. III) COLA’s near-optimal performance.

Experiment Setup The experiments use 4 tabular datasets for binary classification, 5 CE algorithms spanning instance-level, group-level, global, and distributional objectives, and 12 classifiers, as summarized in Table 1. A combination of dataset, A_{CE} algorithm, and predictive model defines one “experiment scenario”.

Table 1: [Experiment Scenarios Setup] Four tabular datasets benchmark five CE algorithms across 12 predictive models. A “scenario” is one combination of dataset, A_{CE} algorithm, and model f .

Dataset	HELOC (FICO, 2018), German Credit (Hofmann, 1994), Hotel Bookings (Antonio et al., 2019), COMPAS (Jeff Larson et al., 2016)
A_{CE}	DiCE (Mothilal et al., 2020), AReS (Rawal & Lakkaraju, 2020), GlobeCE (Ley et al., 2023), KNN (Albini et al., 2022; Contardo et al.; Forel et al., 2023), Discount (You et al., 2025)
Model f	Bagging, LightGBM, Support Vector Machine (SVM), Gaussian Process (GP), Radial Basis Function Network (RBF), XGBoost, Deep Neural Network (DNN), Random Forest (RndForest), AdaBoost, Gradient Boosting (GradBoost), Logistic Regression (LR), Quadratic Discriminant Analysis (QDA)

Table 2: [Experiment Methods Setup] The table defines 6 comparison methods, with colors matching Figure 3. Each method is evaluated within the experiment scenarios defined in Table 1.

Method	The probability $\mathbf{p}$ used by A_{value} and A_{Shap}	The Shapley algorithm A_{Shap}
RB- p_{Uni}	$\mathbf{p} \leftarrow A_{\text{Prob}}: p_{ij} = 1/nm \ (\forall i, j)$	RB-SHAP, $\mathcal{D} = \text{trainset of } f$
RB- p_{OT}	$\mathbf{p} \leftarrow A_{\text{Prob}}: \text{Eq. equation (8) (but not used in } A_{\text{Shap}})$	RB-SHAP, $\mathcal{D} = \text{trainset of } f$
CF- p_{Uni}	$\mathbf{p} \leftarrow A_{\text{Prob}}: p_{ij} = 1/nm \ (\forall i, j)$	CF-SHAP, $\mathcal{D}(\mathbf{x}_i) = \mathbf{p}_i$
CF- p_{Rnd}	$\mathbf{p} \leftarrow A_{\text{Prob}}: \text{Any } \mathbf{x}_i \text{ matched randomly to an } \mathbf{r}_j$	CF-SHAP, $\mathcal{D}(\mathbf{x}_i) = \mathbf{p}_i$
CF- p_{OT}	$\mathbf{p} \leftarrow A_{\text{Prob}}: \text{Eq. equation (8)}$	p -SHAP with $\mathbf{p}$
CF- p_{Ect}	$\mathbf{p} \leftarrow A_{\text{Prob}}: \text{Any } \mathbf{x}_i \text{ matched to known counterpart } \mathbf{r}_j$	(CF or B)-SHAP, $\mathcal{D}(\mathbf{x}_i) \rightarrow \mathbf{r}_j$

We briefly summarize the A_{CE} choices in Table 1. DiCE and KNN are instance-based methods that generate counterfactuals for individual factual instances. AReS and GlobeCE are group-based methods that produce counterfactual instances for the factual data considered as a group. Discount treats the factual instances as an empirical distribution and seeks a nearby counterfactual distribution. Together, these generators cover several common CE scopes while keeping the experimental setup fixed.

Table 2 defines 6 methods: CF- p_{OT} is the proposed p -SHAP instantiation, and the remaining methods are baselines or ablations. Each method is evaluated across the scenarios in Table 1. A method is determined by the combination of A_{Prob} , A_{value} , and A_{Shap} . For instance, RB- p_{Uni} uses a uniform distribution for A_{Prob} to compute $\mathbf{p}$; the resulting $\mathbf{p}$ is then used by A_{value} through equation (9) or equation (10) to compute $\mathbf{q}$, while A_{Shap} is RB-SHAP.

These methods isolate different sources of generalization. First, RB- p_{Uni} and CF- p_{Uni} differ only in whether CE information is used. Second, CF- p_{Uni} , CF- p_{Rnd} , and CF- p_{OT} all use CE information for FA, but with different joint distributions; the results below show that the OT-derived distribution performs substantially better. Third, comparing CF- p_{OT} with RB- p_{OT} tests whether OT helps specifically through factual-counterfactual alignment in A_{Shap} , rather than through other uses of $\mathbf{p}$. Finally, CF- p_{Ect} represents the special case in which each counterfactual has a known source, yielding an exact factual-counterfactual alignment and making CF-SHAP coincide with B-SHAP.

Result I: COLA achieves significant action reduction with a minor loss in counterfactual effect

In Table 3, we set targets requiring $f(\mathbf{z})$ to reach either 80% or 100% of the counterfactual effect of $f(\mathbf{r})$ ². Across these representative scenarios, COLA achieves the target with substantially fewer actions in $\mathbf{z}$, i.e., fewer feature modifications than the original CE $\mathbf{r}$. For the 80% target, the refined explanations use only 13%–25% of the original feature changes as measured by $\|\mathbf{z} - \mathbf{x}\|/\|\mathbf{r} - \mathbf{x}\|$. For the 100% target, only COLA with p -SHAP (CF- p_{OT}) reaches the target, using only 26%–45% of the feature changes in the original $\mathbf{r}$. Thus, within the evaluated datasets, generators, and models, p -SHAP provides the strongest action minimality, which we analyze next.

²Note that by definition, we have $D(f(\mathbf{r}), \mathbf{y}^*) = 0$, which represents a 100% counterfactual effect since $\mathbf{y}^* = f(\mathbf{r})$. To define a counterfactual effect 80%, consider the proportion of divergence reduced by the refined CE $\mathbf{z}$. That is, $\text{Counterfactual Effect} = 1 - D(f(\mathbf{z}), \mathbf{y}^*)/D(f(\mathbf{x}), \mathbf{y}^*) = 80\%$, with D being OT.

Table 3: [COLA for Modification Minimality] This table reports how many features each method modifies in $\mathbf{z}$ when $f(\mathbf{z})$ reaches 80% of the counterfactual effect of $f(\mathbf{r})$. Each entry averages results over 4 randomly selected scenarios from Table 1, with 100 runs per scenario. The symbol “-” indicates that the target counterfactual effect is not achieved.

Dataset	Method	80% Counterfactual Effect		100% Counterfactual Effect	
		# Modified Features	$\frac{\ \mathbf{z}-\mathbf{x}\ }{\ \mathbf{r}-\mathbf{x}\ }$	# Modified Features	$\frac{\ \mathbf{z}-\mathbf{x}\ }{\ \mathbf{r}-\mathbf{x}\ }$
German Credit $ \mathcal{F} = 9$	RB- p_{Uni}	—	—	—	—
	RB- p_{OT}	5.29(± 0.09)	75.9%	—	—
	CF- p_{Uni}	—	—	—	—
	CF- p_{Rnd}	—	—	—	—
	CF- p_{OT}	1.70(± 0.02)	24.3%	3.13(± 0.03)	44.9%
Hotel Bookings $ \mathcal{F} = 29$	RB- p_{Uni}	7.10(± 0.08)	24.5%	—	—
	RB- p_{OT}	8.55(± 0.08)	50.2%	—	—
	CF- p_{Uni}	7.01(± 0.07)	41.1%	—	—
	CF- p_{Rnd}	10.63(± 0.08)	62.4%	—	—
	CF- p_{OT}	2.50(± 0.03)	14.6%	4.44(± 0.02)	26.0%
COMPAS $ \mathcal{F} = 15$	RB- p_{Uni}	5.02(± 0.05)	82.7%	—	—
	RB- p_{OT}	—	—	—	—
	CF- p_{Uni}	2.80(± 0.04)	34.4%	—	—
	CF- p_{Rnd}	2.58(± 0.04)	32.1%	—	—
	CF- p_{OT}	1.25(± 0.03)	14.8%	2.45(± 0.03)	30.0%
HELOC $ \mathcal{F} = 23$	RB- p_{Uni}	—	—	—	—
	RB- p_{OT}	—	—	—	—
	CF- p_{Uni}	—	—	—	—
	CF- p_{Rnd}	2.73(± 0.04)	15.7%	—	—
	CF- p_{OT}	2.35(± 0.03)	13.4%	7.745(± 0.05)	44.7%

Result II: p -SHAP outperforms the other Shapley methods in achieving counterfactual effect

Figure 3 compares the Shapley methods in equation (4)–equation (7), with the action budget C on the x-axis and the objective term $D(f(\mathbf{z}), \mathbf{y}^*)$ from equation (1) on the y-axis. First, RB- p_{Uni} and RB- p_{OT} perform substantially worse than the CE-informed methods, showing that attribution for sparse counterfactual refinement benefits from counterfactual information about $\mathbf{r}$ rather than only background variation around $\mathbf{x}$. Second, CF- p_{OT} outperforms RB- p_{OT} , indicating that OT improves p -SHAP through the contrastive path from $\mathbf{x}$ to $\mathbf{r}$ inside A_{Shap} , not merely through other uses of the joint plan, because these methods differ only in A_{Shap} (see Table 2). Third, p -SHAP substantially outperforms CF- p_{Uni} and CF- p_{Rnd} , showing that using counterfactual information alone is not sufficient for constructing an effective $\mathbf{z}$. The joint distribution induced by OT provides a more effective alignment between factual and counterfactual data, even though this alignment need not equal the exact generation mechanism, as further illustrated in Figure 4.

Overall, COLA with p -SHAP performs well across many scenarios in Table 1, as shown in Figure 3 and Table 3. Additional scenario-level results appear in Appendix H. Within these evaluated conditions, p -SHAP outperforms the other Shapley variants for sparse CE refinement.

Result III: COLA is competitive with the mixed integer linear programming (MILP) optimum on tractable benchmarks

This result evaluates whether p -SHAP can reduce dependence on CE-generator-specific alignment knowledge³ by replacing it with the OT joint distribution between factual and counterfactual data, as shown in Figure 4. We benchmark CF- p_{Ect} within COLA and solve equation (1) with MeanD as the divergence function D . Because CF- p_{Ect} relies on known factual-counterfactual alignment, this comparison tests how effectively COLA can exploit such alignment when it is available. The optimal solution for this tractable case can be obtained through an MILP; Appendix G derives the formulation. Since solving the MILP is computationally expensive, this benchmark is run only for *German Credit*.

³This knowledge, i.e. exact alignment between factual and counterfactual, is available only by DiCE and KNN among all CE methods considered.

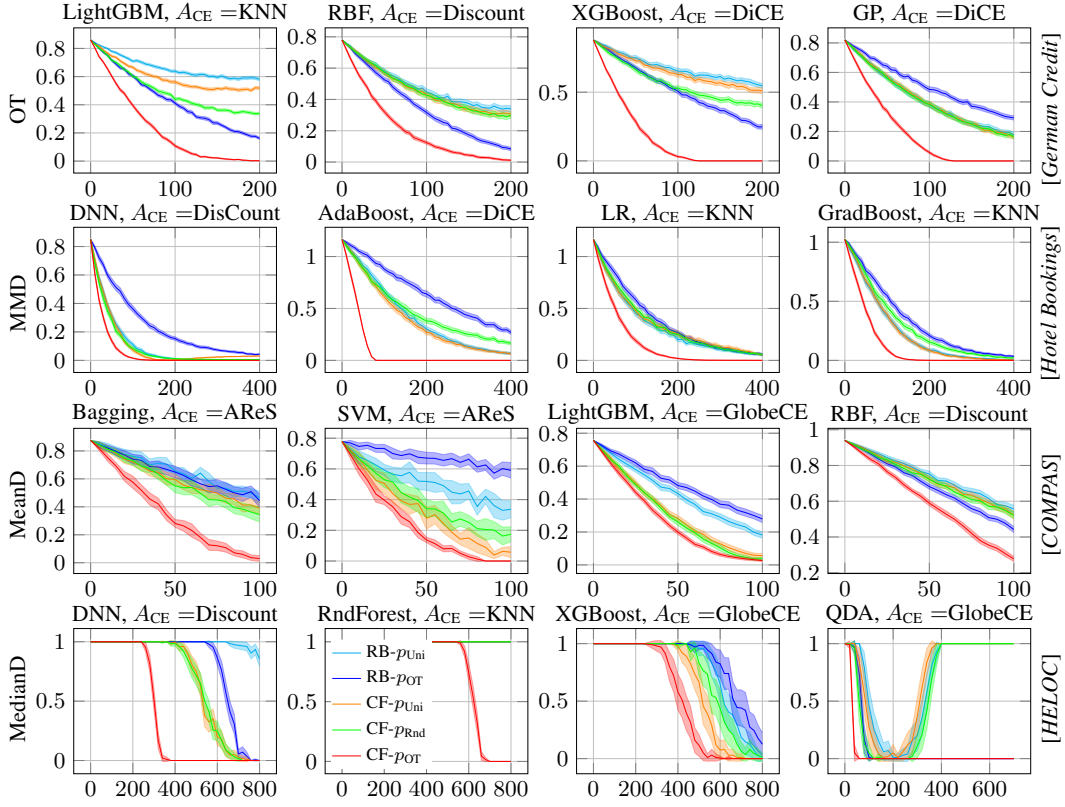


Figure 3: $D(f(z), y^*)$ as a function of the action budget C across the evaluated scenarios. Each experiment uses 100 runs, and shaded regions indicate 99.9% confidence intervals. $A_{\text{Value}}^{\text{avg}}$ is used for *HELOC* and *COMPAS*, while $A_{\text{Value}}^{\text{max}}$ is used for *German Credit* and *Hotel Bookings*. The legend inside “RnDForest, $A_{\text{CE}} = \text{KNN}$ ” applies to all plots.

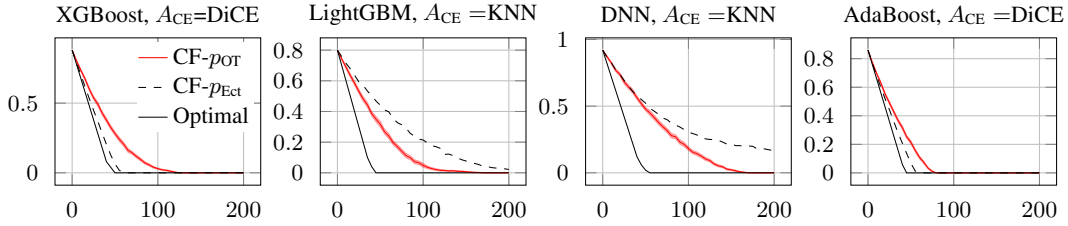


Figure 4: [*German Credit*] $D(f(z), y^*)$ as a function of the action budget C , using *MeanD* as D .

$\text{CF-}p_{\text{Ect}}$ is near-optimal when used with DiCE. For DiCE and KNN, factual-counterfactual pairs are independent across instances, so these results indicate that COLA is effective for instance-based CE even though equation (1) also covers group and distributional CE settings. $\text{CF-}p_{\text{OT}}$ sometimes outperforms $\text{CF-}p_{\text{Ect}}$, because it utilizes a more theoretically grounded approach to identify the key features that require modification, whereas $\text{CF-}p_{\text{Ect}}$ relies on CE algorithm-dependent knowledge, which lacks solid justification on its effectiveness for FA. A gap remains between $\text{CF-}p_{\text{Ect}}$, $\text{CF-}p_{\text{OT}}$, and the optimum for KNN, indicating that identifying the best alignment remains an open question.

7 CONCLUSIONS

This paper introduces COLA, a model- and generator-agnostic framework for refining CE with joint-distribution-informed Shapley values. Across the evaluated tabular datasets, classifiers, CE generators, and divergence choices, COLA preserves counterfactual effects with substantially fewer

actions, while the theoretical analysis explains why OT-based alignment is an effective general mechanism for attribution-driven refinement.

ETHICS STATEMENT

All authors have read and agree to abide by the ICLR Code of Ethics.⁴ This paper proposes a broadly applicable refinement method (*COLA* with *p-SHAP*) for existing CE outputs. The work is methodological and empirical; it does not involve human subjects or interventions, and we did not collect, annotate, or release any new personal data.

All experiments use standard, de-identified, publicly available tabular datasets (e.g., Adult/Census Income, German Credit, HELOC) as benchmarks for the proposed refinement framework. We follow dataset licenses/terms and common research practice by using these data solely for benchmarking. We do not attempt re-identification or linkage, and we release no additional personal information. Our code will include options to exclude protected attributes and to enforce feasibility/immutability constraints on features.

CE can surface biases in underlying models, but neither CE nor our refinement step guarantees fairness. COLA can reduce the number of recommended feature edits; it does *not* authorize edits to sensitive or immutable attributes (e.g., sex, race, age) and does not confer causal validity. We discourage using CE to “game” high-stakes systems (e.g., lending, hiring, healthcare) or to pressure individuals into unrealistic or unethical behavior changes. Any deployment in consequential domains should add domain-expert-defined feasibility constraints, fairness audits (e.g., disparate impact/benefit across groups), human oversight, and compliance with applicable data-protection and AI regulations. We explicitly caution that our attribution is distributional, not causal, and must not be interpreted as causal effect.

Risks include (i) misinterpreting attributions as causal, (ii) recommending infeasible or harmful edits, and (iii) enabling strategic manipulation across application domains. We mitigate these risks by (a) stating the non-causal scope and limitations of the method, (b) supporting cost/feasibility masks and immutable features in the implementation, (c) reporting performance across subgroups where relevant, and (d) providing responsible-use guidance in documentation.

We provide an anonymized implementation, detailed settings, and seeds to facilitate replication across the same datasets, models, and CE generators studied here. We report model types, hyperparameters, and compute budgets, and avoid result cherry-picking by using fixed evaluation protocols. To our knowledge, there are no conflicts of interest or sensitive sponsorships related to this submission; any funding disclosures will be provided after the double-blind review.

REPRODUCIBILITY STATEMENT

We take reproducibility seriously and provide pointers to all components needed to reproduce the reported scope of experiments. The COLA framework and *p-SHAP* attribution are specified with pseudocode and notation in the main paper (Algorithm 1; Method and Preliminaries sections), and the theoretical assumptions and complete proofs are provided in the appendix through theorem and lemma statements with line-by-line proofs.

Implementation details—hyperparameters, optimization settings, environment specifications, and random seeds—are documented in Appendix I, which also describes the datasets and evaluation protocol, including splits and preprocessing. Baselines, including predictive models and CE generators, as well as coupling choices are enumerated in the experimental sections and ablations, with matching configurations in the supplemental materials. The experiment code and software will be released upon acceptance; Appendix I reports hardware and runtime to contextualize compute requirements, and the appendix provides ablation/sensitivity studies plus the small-scale MILP benchmark setup for independent verification.

⁴<https://iclr.cc/public/CodeOfEthics>

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A COMPARISON TO EXISTING APPROACHES

CF-SHAP (Albini et al., 2022) uses counterfactual data points as the background distribution for Shapley attribution. Its scope, however, assumes that the counterfactual distribution is defined conditionally for each individual factual instance, which implies a known probabilistic alignment between factual instances and counterfactual $\mathbf{r}$ instances; related assumptions appear in (Merrick & Taly, 2020; Kommiya Mothilal et al., 2021). This instance-conditioned setup in (Albini et al., 2022) is a special case of our formulation and can be restrictive when explanations are global, group-based, or distributional. WeightedSHAP (Kwon & Zou, 2022) changes the feature-weighting scheme rather than treating features as equally contributing, whereas our method weights the contribution of data-point alignments and can be extended to combine row and column weighting. The framework in (Kommiya Mothilal et al., 2021) also uses both FA and CE, but its CE-based attribution inherits the same conditional-alignment assumption as (Albini et al., 2022), making it less directly applicable to group (Ley et al., 2023; 2022) or distributional CE (You et al., 2025). Another relevant line, represented by (Sharma et al.), uses causal structure and time-series predictors to decompose a single observed system-level change into input contributions. These works address important attribution and counterfactual questions, but they do not directly study the minimal-action CE refinement problem considered here.

The problems formulated in (Kanamori et al., 2022; Karimi et al., 2021) are closer to our action-minimality objective. In (Karimi et al., 2021), the action cost is minimized under the assumption of a known structural causal model (SCM), which is rarely available in practice. (Kanamori et al., 2022) notes that it remains open whether existing CE methods can be used to solve the action-selection problem, and proposes an MILP-solver-based approach for linear classifiers, tree ensembles, and deep ReLU networks, building on (Ustun et al., 2019; Kanamori et al., 2020; Parmentier & Vidal, 2021). Because MILP solving is expensive, such exact approaches are difficult to scale broadly, motivating a generator-agnostic post-hoc refinement framework.

B $\mathcal{NP}$ -HARDNESS OF THE PROBLEM IN EQUATION (1)

Theorem B.1 shows that a scalable exact algorithm for equation (1) should not be expected in general unless $\mathcal{P} = \mathcal{NP}$. The difficulty is not limited to nonlinear models; it also arises from the combinatorial action-selection structure.

Theorem B.1. *Problem equation (1) is generally $\mathcal{NP}$ -hard for non-trivial divergences. More specifically, it is hard even when $d = 1$ for linear models.*

Proof. Consider the *Sparse Regression (SR) with a Cardinality Constraint* problem, a standard sparse-selection problem. Given a matrix $\mathbf{W} \in \mathbb{R}^{m \times n}$, a target vector $\mathbf{y}^* \in \mathbb{R}^m$, and a sparsity level $K \in \mathbb{N}$, the task is to find a vector $\mathbf{z} \in \mathbb{R}^n$ that minimizes residual error while using at most K

non-zero entries:

$$\begin{aligned} \min_{\mathbf{z}} \quad & \|\mathbf{W}\mathbf{z} - \mathbf{y}^*\|_2^2, \\ \text{s.t.} \quad & \|\mathbf{z}\|_0 \leq K, \end{aligned}$$

where $\|\mathbf{z}\|_0$ denotes the number of non-zero elements in $\mathbf{z}$. This problem is known to be $\mathcal{NP}$ -hard due to the combinatorial nature of selecting the subset of variables to include in the model.

We map this SR problem to Problem equation (1) under a deliberately simple setting, which makes the hardness result broadly informative. Let the number of features be $d = 1$ and the number of instances be n , matching the dimension of the SR problem. Let the factual data be $\mathbf{x} = \mathbf{0} \in \mathbb{R}^n$, and let the target output be the same $\mathbf{y}^* \in \mathbb{R}^m$. Define the model f as the linear function $f(\mathbf{z}) = \mathbf{W}\mathbf{z}$.

For the model output, we define $D(f(\mathbf{z}), \mathbf{y}^*) = \|\mathbf{W}\mathbf{z} - \mathbf{y}^*\|_2^2$, i.e. the Euclidean distance squared. For the instances, we define $D(\mathbf{z}, \mathbf{x}) = \|\mathbf{z} - \mathbf{x}\|_2^2 = \|\mathbf{z}\|_2^2$, since $\mathbf{x} = \mathbf{0}$. We set the maximum allowed feature changes C to be equal to k (the sparsity level from the SR problem). The large constant M can be any sufficiently large positive number, for example, $M \geq \max_i |z_i|$.

Given that $\mathbf{x} = \mathbf{0}$, constraints equation (1d) and equation (1e) are simplified to:

$$\begin{aligned} z_i &\leq 0 \cdot (1 - c_i) + M c_i = M c_i, \\ z_i &\geq 0 \cdot (1 - c_i) - M c_i = -M c_i, \quad \forall k = 1, \dots, n. \end{aligned}$$

This means that if $c_i = 0$, then $z_i \leq 0$ and $z_i \geq 0$, so $z_i = 0$. If $c_i = 1$, then $z_i \in [-M, M]$. The constraint equation (1c) becomes $\sum_{k=1}^n c_i \leq k$.

Our problem equation (1) thus becomes:

$$\min_{\mathbf{c}, \mathbf{z}} \quad \|\mathbf{W}\mathbf{z} - \mathbf{y}^*\|_2^2, \tag{11a}$$

$$\text{s.t.} \quad \|\mathbf{z}\|_2^2 \leq \epsilon, \tag{11b}$$

$$\sum_{k=1}^d c_i \leq k, \tag{11c}$$

$$z_i = 0, \quad \text{if } c_i = 0, \tag{11d}$$

$$z_i \in [-M, M], \quad \text{if } c_i = 1, \tag{11e}$$

$$c_i \in \{0, 1\}, \quad \forall k = 1, \dots, n. \tag{11f}$$

In this formulation, the variables c_i indicate whether the variable z_i is allowed to change ($c_i = 1$) or not ($c_i = 0$). The constraints enforce that $z_i = 0$ when $c_i = 0$, mirroring the sparsity constraint in the SR problem. The constraint $\sum_{k=1}^n c_i \leq k$ ensures that at most k features can change, matching the sparsity level. The objective function is identical to that of SR.

Because this construction directly mirrors cardinality-constrained SR, which is $\mathcal{NP}$ -hard, solving Problem equation (1) is at least as hard as solving the sparse regression problem. Therefore, Problem equation (1) remains $\mathcal{NP}$ -hard even in the restricted case $d = 1$, with a linear model f and standard Euclidean divergences. $\square$

C PROOF OF THEOREM 4.1: p -SHAP TOWARDS COUNTERFACTUAL EFFECT

Let $\mathbf{x} = \{\mathbf{x}_i\}_{i=1}^n \in \mathbb{R}^{n \times d}$ be the set of factual data points with associated probability weights $\mu_i \geq 0$ which implies $\sum_{i=1}^n \mu_i = 1$, and let $\mathbf{r} = \{\mathbf{r}_j\}_{j=1}^m \in \mathbb{R}^{m \times d}$ be the set of counterfactual data points with associated probability weights $\nu_j \geq 0$ which implies $\sum_{j=1}^m \nu_j = 1$. Let $\mathbf{p}_{\text{OT}} \in \mathbb{R}^{n \times m}$ be the OT plan between $\mathbf{x}$ and $\mathbf{r}$ that minimizes the expected transportation cost:

$$\mathbf{p}_{\text{OT}} = \arg \min_{\mathbf{p} \in \Pi(\boldsymbol{\mu}, \boldsymbol{\nu})} \sum_{i=1}^n \sum_{j=1}^m p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2,$$

where $\Pi(\boldsymbol{\mu}, \boldsymbol{\nu})$ is the set of joint distributions satisfying the marginal constraints $\sum_{j=1}^m p_{ij} = \mu_i$ for all i and $\sum_{i=1}^n p_{ij} = \nu_j$ for all j . The theorem below formalizes how the OT plan aligns attribution with expected feature-modification costs, supporting action plans that are cost-efficient for achieving counterfactual outcomes across aligned factual-counterfactual sets.

Theorem C.1 (Theorem 4.1 in the main text). *Consider the 1-Wasserstein divergence W_1 , i.e. $W_1(f(\mathbf{x}), \mathbf{y}^*) = \min_{\pi \in \Pi} \sum_{i=1}^n \sum_{j=1}^m \pi_{ij} |f(\mathbf{x}_i) - \mathbf{y}_j^*|$. Suppose the counterfactual outcome $\mathbf{y}^*$ is fully achieved by $\mathbf{r}$, i.e. $\mathbf{y}_j^* = f(\mathbf{r}_j)$ ($j = 1, 2, \dots, m$). Assume that the model $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is Lipschitz continuous with Lipschitz constant L . The expected absolute difference in model outputs between the factual and counterfactual instances, weighted by $\mathbf{p}_{OT}$ (with $\varepsilon = 0$), is bounded by:*

$$W_1(f(\mathbf{x}), \mathbf{y}^*) \leq L \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2} \leq L \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2} \quad \forall \mathbf{p} \in \Pi.$$

Thus, $\mathbf{p}_{OT}$ minimizes the expected-modification-cost upper bound on $W_1(f(\mathbf{x}), \mathbf{y}^*)$.

Proof. Because the model f is assumed Lipschitz continuous with constant L , the following bound holds for any factual point $\mathbf{x}_i \in \mathbb{R}^d$ and counterfactual point $\mathbf{r}_j \in \mathbb{R}^d$:

$$|f(\mathbf{x}_i) - f(\mathbf{r}_j)| \leq L \|\mathbf{x}_i - \mathbf{r}_j\|_2.$$

Multiplying both sides by the nonnegative OT weight $p_{ij}^{OT} \geq 0$ gives:

$$p_{ij}^{OT} |f(\mathbf{x}_i) - f(\mathbf{r}_j)| \leq L p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2.$$

Summing this inequality over all factual-counterfactual pairs $i = 1, \dots, n$ and $j = 1, \dots, m$ yields:

$$\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} |f(\mathbf{x}_i) - f(\mathbf{r}_j)| \leq L \sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2.$$

Let $E_f = \sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} |f(\mathbf{x}_i) - f(\mathbf{r}_j)|$ and $E_d = \sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2$. In this notation, the bound is:

$$E_f \leq L E_d.$$

To bound E_d , we apply the Cauchy-Schwarz inequality. The weights p_{ij}^{OT} are nonnegative and satisfy $\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} = 1$, because p_{OT} is a probability distribution over the joint factual-counterfactual space. For real-valued arrays a_{ij} and b_{ij} , Cauchy-Schwarz gives

$$\left(\sum_{i,j} a_{ij} b_{ij} \right)^2 \leq \left(\sum_{i,j} a_{ij}^2 \right) \left(\sum_{i,j} b_{ij}^2 \right).$$

Setting $a_{ij} = \sqrt{p_{ij}^{OT}}$ and $b_{ij} = \sqrt{p_{ij}^{OT}} \|\mathbf{x}_i - \mathbf{r}_j\|_2$, we have:

$$E_d = \sum_{i,j} p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2 = \sum_{i,j} \sqrt{p_{ij}^{OT}} \sqrt{p_{ij}^{OT}} \|\mathbf{x}_i - \mathbf{r}_j\|_2 = \sum_{i,j} a_{ij} b_{ij}.$$

Applying the inequality gives:

$$E_d^2 \leq \left(\sum_{i,j} a_{ij}^2 \right) \left(\sum_{i,j} b_{ij}^2 \right) = \left(\sum_{i,j} p_{ij}^{OT} \right) \left(\sum_{i,j} p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2 \right).$$

Because $\sum_{i,j} p_{ij}^{OT} = 1$, this simplifies to:

$$E_d^2 \leq \sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2 = \sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2.$$

Taking square roots gives:

$$E_d \leq \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2}.$$

Substituting this bound back into the inequality for E_f yields:

$$E_f \leq L \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{\text{OT}} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2}.$$

The 1-Wasserstein divergence is no larger than E_f ⁵; therefore,

$$W_1(f(\mathbf{x}), \mathbf{y}^*) = \min_{\pi \in \Pi} \sum_{i=1}^n \sum_{j=1}^m \pi_{ij} |f(\mathbf{x}_i) - \mathbf{y}_j^*| \leq E_f \leq L \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{\text{OT}} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2}.$$

Thus, the 1-Wasserstein divergence between $f(\mathbf{x})$ and $\mathbf{y}^*$ is bounded by the Lipschitz constant L times the square root of the expected transportation cost under $\mathbf{p}_{\text{OT}}$. Since $\mathbf{p}_{\text{OT}}$ minimizes $\sum_{i,j} p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2$ over all feasible transport plans in $\Pi(\boldsymbol{\mu}, \boldsymbol{\nu})$,

$$\sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{\text{OT}} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2} \leq \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2}, \quad \forall \mathbf{p} \in \Pi(\boldsymbol{\mu}, \boldsymbol{\nu}).$$

The stated bound follows. $\square$

D PROOF OF THEOREM 4.2: INTERVENTIONAL EFFECT

Theorem D.1 (Theorem 4.2 in the main text). *For any subset $S \subseteq \mathcal{F}$ and any factual instance $\mathbf{x}_i$ ($i = 1, 2, \dots, n$), $v^{(i)}(S)$ represents the difference between the expected value of $f(\mathbf{r})$ under an intervention on features S and the unconditional expected value of $f(\mathbf{r})$. This interventional interpretation is expressed as:*

$$\mathbb{E}[f(\mathbf{r})] + v^{(i)}(S) = \mathbb{E}[f(\mathbf{r}) | do(\mathbf{r}_S = \mathbf{x}_{i,S})].$$

Proof. Let $p(\mathbf{x}, \mathbf{r})$ be the joint probability over factual and counterfactual data returned by A_{prob} . Under the intervention $do(\mathbf{r}_S = \mathbf{x}_{i,S})$, the features in S are fixed to $\mathbf{x}_{i,S}$, while the remaining features in $\mathcal{F} \setminus S$ follow their marginal distribution $p(\mathbf{r}_{\mathcal{F} \setminus S})$. Thus, the expected value of $f(\mathbf{r})$ under this intervention is:

$$\mathbb{E}[f(\mathbf{r}) | do(\mathbf{r}_S = \mathbf{x}_{i,S})] = \int_{\mathcal{R}_{\mathcal{F} \setminus S}} f(\mathbf{x}_{i,S}; \mathbf{r}_{\mathcal{F} \setminus S}) p(\mathbf{r}_{\mathcal{F} \setminus S}) d\mathbf{r}_{\mathcal{F} \setminus S}.$$

By the definition of $v^{(i)}(S)$ in the joint-probability-informed attribution,

$$v^{(i)}(S) = \mathbb{E}_{\mathbf{r} \sim p(\mathbf{r} | \mathbf{x}_i)}[f(\mathbf{x}_{i,S}; \mathbf{r}_{\mathcal{F} \setminus S})] - \mathbb{E}[f(\mathbf{r})] = \int_{\mathcal{R}_{\mathcal{F} \setminus S}} f(\mathbf{x}_{i,S}; \mathbf{r}_{\mathcal{F} \setminus S}) p(\mathbf{r}_{\mathcal{F} \setminus S}) d\mathbf{r}_{\mathcal{F} \setminus S} - \mathbb{E}[f(\mathbf{r})],$$

which implies

$$\mathbb{E}[f(\mathbf{r})] + v^{(i)}(S) = \mathbb{E}[f(\mathbf{r}) | do(\mathbf{r}_S = \mathbf{x}_{i,S})].$$

This proves the stated interventional identity. $\square$

The value function $v^{(i)}(S)$ therefore describes how the expected model output changes when the counterfactual features in S are replaced by $\mathbf{x}_{i,S}$, denoted by $do(\mathbf{r}_S = \mathbf{x}_{i,S})$. This interpretation is tied to the intervention represented by the attribution term, rather than to a claim that the empirical coupling is a full causal model. Consequently, $\mathbb{E}[f(\mathbf{r})] + v^{(i)}(S)$ combines the baseline expected counterfactual output with the additional interventional contribution associated with replacing the selected features.

⁵Note that $E_f = \sum_{i=1}^n \sum_{j=1}^m p_{ij}^{\text{OT}} |f(\mathbf{x}_i) - f(\mathbf{r}_j)|$. Because both π and $\mathbf{p}_{\text{OT}}$ denote joint probability of $\mathbf{x}$ and $\mathbf{r}$, however, π makes the summation the minimum for the W_1 term across all possible joint distributions, whereas $\mathbf{p}_{\text{OT}}$ does not.

E PROOF OF THEOREM 5.1: COUNTERFACTUAL PROXIMITY

Theorem E.1 (Theorem 5.1 in the main text). *Let $\mathbf{q} \in \mathbb{R}^{n \times d}$ be the aligned reference matrix derived from the counterfactual set $\mathbf{r} \in \mathbb{R}^{m \times d}$ via the value assignment step $\mathbf{q} = A_{\text{value}}(\mathbf{r}, \mathbf{p})$, where $\mathbf{p}$ is the coupling matrix. For any dimensions $n, m \geq 1$, the refined counterfactual $\mathbf{z}$ constructed by the COLA framework satisfies:*

$$\|\mathbf{z} - \mathbf{x}\|_F \leq \|\mathbf{q} - \mathbf{x}\|_F.$$

This inequality guarantees that the refined counterfactual $\mathbf{z}$ is strictly closer (or equal) to the factual input $\mathbf{x}$ compared to the aligned counterfactual proposal $\mathbf{q}$, ensuring that the refinement process introduces no additional divergence.

Proof. Let $\mathbf{q}$ be the aligned reference matrix generated by $A_{\text{value}}(\mathbf{r}, \mathbf{p})$. For the elements where $c_{ik} = 1$ (modified elements), $z_{ik} = q_{ik}$. For elements where $c_{ik} = 0$, $z_{ik} = x_{ik}$. Therefore, we can write:

$$(z_{ik} - x_{ik})^2 = \begin{cases} (q_{ik} - x_{ik})^2, & \text{if } c_{ik} = 1; \\ 0, & \text{if } c_{ik} = 0. \end{cases} \quad (12)$$

The squared Frobenius norms with respect to $\mathbf{q}$ and $\mathbf{z}$ are computed as follows:

$$\begin{aligned} \|\mathbf{q} - \mathbf{x}\|_F^2 &= \sum_{i=1}^n \sum_{k=1}^d (q_{ik} - x_{ik})^2, \\ \|\mathbf{z} - \mathbf{x}\|_F^2 &= \sum_{i=1}^n \sum_{k=1}^d (z_{ik} - x_{ik})^2. \end{aligned}$$

And,

$$\begin{aligned} \|\mathbf{q} - \mathbf{x}\|_F^2 - \|\mathbf{z} - \mathbf{x}\|_F^2 &= \sum_{i=1}^n \sum_{k=1}^d (q_{ik} - x_{ik})^2 - \sum_{i=1}^n \sum_{k=1}^d (z_{ik} - x_{ik})^2 \\ &\stackrel{(i)}{=} \sum_{i=1}^n \sum_{k=1}^d (q_{ik} - x_{ik})^2 - \sum_{(i,k): c_{ik}=1} (q_{ik} - x_{ik})^2 \\ &= \sum_{(i,k): c_{ik}=0} (q_{ik} - x_{ik})^2 \geq 0, \end{aligned}$$

where the equality (i) holds because of equation (12).

Since the difference $\|\mathbf{q} - \mathbf{x}\|_F^2 - \|\mathbf{z} - \mathbf{x}\|_F^2 \geq 0$, it follows that:

$$\|\mathbf{z} - \mathbf{x}\|_F^2 \leq \|\mathbf{q} - \mathbf{x}\|_F^2.$$

Taking square roots, we get:

$$\|\mathbf{z} - \mathbf{x}\|_F \leq \|\mathbf{q} - \mathbf{x}\|_F.$$

This proves the stated interventional identity. $\square$

Corollary E.2 (Corollary 5.2 in the main text). *In the special case where $n = m$ and the OT plan $\mathbf{p}$ corresponds to a deterministic permutation σ (i.e., obtained without entropic regularization, $\varepsilon = 0$), the matrix $\mathbf{q}$ becomes a row-permuted version of $\mathbf{r}$ (denoted as $\mathbf{r}_\sigma$, where $q_i = r_{\sigma(i)}$). In this setting, the bound simplifies to:*

$$\|\mathbf{z} - \mathbf{x}\|_F \leq \|\mathbf{r}_\sigma - \mathbf{x}\|_F,$$

recovering the intuition that the refined counterfactual is at least as close to the factuals as the original counterfactual set after reordering by σ .

Proof. Under $n = m$ and $\varepsilon = 0$, the optimal transport plan $\mathbf{p}$ is a permutation matrix associated with a bijection σ . Therefore, the value-assignment step A_{value} , whether maximizing or averaging, assigns $\mathbf{q}_i = \mathbf{r}_{\sigma(i)}$ for every instance i . Replacing $\mathbf{q}$ with the permuted matrix $\mathbf{r}_\sigma$ in the general bound from Theorem 5.1, namely $\|\mathbf{z} - \mathbf{x}\|_F \leq \|\mathbf{q} - \mathbf{x}\|_F$, gives the result. $\square$

F ANALYSIS OF COMPUTATIONAL COMPLEXITY OF COLA

We analyze the computational complexity of COLA by separating the generator, alignment, attribution, value-assignment, and action-selection components.

First, consider A_{Prob} . If the alignment between $\mathbf{x}$ and $\mathbf{r}$ is known a priori, then A_{Prob} only constructs the matrix $\mathbf{p}$ from that knowledge, taking $\mathcal{O}(n \times m)$. Otherwise, we solve OT to obtain $\mathbf{p}$; with the Sinkhorn–Knopp algorithm, this takes $\mathcal{O}(n \times m \times \log(1/\varepsilon))$.

Next consider A_{Shap} . For each sampled subset, the model is evaluated on all n data points, giving $\mathcal{O}(n)$ evaluations per subset. Incorporating reference values from $\mathbf{r}$ requires replacing selected feature values with corresponding baseline values. This operation can require $\mathcal{O}(m)$ access to the reference table, but with precomputed reference values it is treated as $\mathcal{O}(d)$. With M_{Shap} sampled subsets, the resulting complexity for A_{Shap} is $\mathcal{O}(n \times d \times M_{\text{Shap}})$.

The normalization in line 4 of COLA takes $\mathcal{O}(n \times d)$.

To compare the complexities of the two algorithms, $A_{\text{Value}}^{\text{max}}$ and $A_{\text{Value}}^{\text{avg}}$, we analyze each algorithm step-by-step. For $A_{\text{Value}}^{\text{max}}$, for each row $\mathbf{x}_i$ in the data table, we need to (1) compute the probabilities p_{ij} for all $j \in \{1, 2, \dots, m\}$, which involves $\mathcal{O}(m)$ operations per row, (2) identify the row $\mathbf{r}_j$ in the reference data with the highest probability p_{ij} , which involves $\mathcal{O}(m)$ operations per row, and (3) assign $q_{ik} = r_{\tau(i),k}$ where $\tau(i) = \arg \max_j p_{ij}$, which involves $\mathcal{O}(d)$ operations per row. Since there are n rows in the data table, the total complexity for $A_{\text{Value}}^{\text{max}}$ is $\mathcal{O}(n \times (m + m + d)) = \mathcal{O}(n \times (2m + d)) = \mathcal{O}(n \times m + n \times d) = \mathcal{O}(n \times (m + d))$.

For $A_{\text{Value}}^{\text{avg}}$, for each row $\mathbf{x}_i$ in the data table, we need to 1) compute the probabilities p_{ij} for all $j \in \{1, 2, \dots, m\}$, which involves $\mathcal{O}(m)$ operations per row, 2) compute the sum $\sum_{j'=1}^m p_{ij'}$, which involves $\mathcal{O}(m)$ operations per row, and 3) calculate the weighted average $q_{ik} = \sum_{j=1}^m \left(\frac{p_{ij}}{\sum_{j'=1}^m p_{ij'}} \right) r_{jk}$, which involves $\mathcal{O}(m \times d)$ operations per row. Since there are n rows in the data table, the total complexity for $A_{\text{Value}}^{\text{avg}}$ is $\mathcal{O}(n \times (m + m + m \times d)) = \mathcal{O}(n \times (2m + m \times d)) = \mathcal{O}(n \times (m + m \times d)) = \mathcal{O}(n \times m \times (1 + d)) = \mathcal{O}(n \times m \times d)$.

For lines 6–16, the entire complexity is straightforwardly $\mathcal{O}(n \times d) + \mathcal{O}(C) = \mathcal{O}(n \times d)$ due to the fact $C \leq n \times d$.

Therefore, the complexity of COLA using $A_{\text{Value}}^{\text{max}}$ equals

$$\begin{aligned} & \mathcal{O}(M_{\text{CE}}) + \mathcal{O}(nm \log(1/\varepsilon)) + \mathcal{O}(ndM_{\text{Shap}}) + \mathcal{O}(nd) + \mathcal{O}(n(m + d)) + \mathcal{O}(nd) \\ &= \mathcal{O}(M_{\text{CE}}) + \mathcal{O}(nm \log(1/\varepsilon)) + \mathcal{O}(ndM_{\text{Shap}}) + \mathcal{O}(nm) + \mathcal{O}(nd), \end{aligned}$$

and the complexity of COLA using $A_{\text{Value}}^{\text{avg}}$ equals

$$\begin{aligned} & \mathcal{O}(M_{\text{CE}}) + \mathcal{O}(nm \log(1/\varepsilon)) + \mathcal{O}(ndM_{\text{Shap}}) + \mathcal{O}(nd) + \mathcal{O}(nmd) + \mathcal{O}(nd) \\ &= \mathcal{O}(M_{\text{CE}}) + \mathcal{O}(nm \log(1/\varepsilon)) + \mathcal{O}(ndM_{\text{Shap}}) + \mathcal{O}(nmd). \end{aligned}$$

Combining these components, the complexity of COLA with respect to n , m , d , and the entropic OT regularization parameter ε is

$$\mathcal{O}(M_{\text{CE}}) + \mathcal{O}(nm \log(1/\varepsilon)) + \mathcal{O}(ndM_{\text{Shap}}) + N,$$

where $N = \mathcal{O}(nm) + \mathcal{O}(nd)$ if $A_{\text{Value}}^{\text{max}}$ is used and $N = \mathcal{O}(nmd)$ if $A_{\text{Value}}^{\text{avg}}$ is used.

G AN MILP FORMULATION OF EQUATION (1) WITH MEAND

This section provides a global optimality benchmark for the tractable case where factual-counterfactual alignment is known and equation (1) is solved with D equal to MeanD:

$$D(f(\mathbf{z}), \mathbf{y}^*) = \left| \frac{1}{n} \sum_{i=1}^n f(\mathbf{z}_i) - \bar{y}^* \right|,$$

with $\bar{y}^* = \frac{1}{m} \sum_{j=1}^m y_j^*$. Because COLA starts from $D(\mathbf{r}, \mathbf{x}) \leq \epsilon$ and constructs $\mathbf{z}$ no farther from $\mathbf{x}$ than the aligned counterfactual proposal, equation (1b) can be omitted in this benchmark. The

resulting formulation is:

$$\min_{\mathbf{c}, \mathbf{z}} \left| \sum_{i=1}^n f(\mathbf{z}_i) - ny^* \right|, \quad (13a)$$

$$\text{s.t.} \quad \sum_{i=1}^n \sum_{k=1}^d c_{ik} \leq C, \quad (13b)$$

$$z_{ik} = r_{ik}c_{ik} + x_{ik}(1 - c_{ik}), \quad i = 1, \dots, n, \quad k = 1, \dots, d. \quad (13c)$$

Note that the original constraints equation (1d) and equation (1e) merge to be equation (13c), because CF- p_{Ect} is imposed to be used. That is, for any $\mathbf{x}_i$, there is an exact $\mathbf{r}_j$ serves as its reference in A_{Shap} and A_{Value} , which implies x_{ik} ($k = 1, 2, \dots, d$) either stays unchanged or can be changed to r_{jk} . Therefore $z_{ik} = r_{ik}c_{ik} + x_{ik}(1 - c_{ik})$ of which the value depends on the binary variable c_{ik} .

Due to the known alignment between any $\mathbf{x}_i$ and its corresponding $\mathbf{r}_j$, $\mathbf{q}$ is determined (also, both $A_{\text{Values}}^{\text{max}}$ and $A_{\text{Values}}^{\text{avg}}$ return the same $\mathbf{q}$). For any data point i and any feature set $S \subseteq \mathcal{F}$, let $\mathbf{z}_{iS}$ denotes the solution $\mathbf{z}_i$ where we have all features $k \in S$ changed to q_{ik} , and the other features $h \in \mathcal{F} \setminus S$ stays x_{ih} . Hence the set of $\mathbf{z}_{iS}$ ($S \subseteq \mathcal{F}$) composes the domain of all possible values of $\mathbf{z}$. Define a corresponding scaler variable for any $\mathbf{z}_{iS}$:

$$g_{iS} = f(\mathbf{z}_{iS}) - \bar{y}^*.$$

Then, for any $\mathbf{z}_i$ in equation (13), the value of the term $\sum_{i=1}^n f(\mathbf{z}_i) - ny^*$ can be represented by a binary variable a_{iS} together with the scaler g_{iS} , namely,

$$\sum_{i=1}^n f(\mathbf{z}_i) - ny^* = \sum_{i=1}^n \sum_{S \subseteq \mathcal{F}} g_{iS} a_{iS}.$$

The optimization problem equation (13) is hence reformulated as a mixed integer programming below.

$$\min_{\mathbf{a}, \eta} \quad \eta, \quad (14a)$$

$$\text{s.t.} \quad \sum_{i=1}^n \sum_{S \subseteq \mathcal{F}} g_{iS} a_{iS} \leq \eta, \quad (14b)$$

$$\sum_{i=1}^n \sum_{S \subseteq \mathcal{F}} g_{iS} a_{iS} \geq -\eta, \quad (14c)$$

$$\sum_{S \subseteq \mathcal{F}} a_{iS} = 1, \quad i = 1, \dots, n, \quad (14d)$$

$$\sum_{i=1}^n \sum_{S \subseteq \mathcal{F}} |S| a_{iS} \leq C. \quad (14e)$$

Minimizing η under the two constraints equation (14b) and equation (14c) is equivalent to minimizing the objective function of equation (13). The constraints in equation (14d) guarantees that each data point i is subject to one and only one feature modification plan a_{iS} for a specific S ($S \subseteq \mathcal{F}$). The constraint equation (14e) corresponds to equation (13b). Solving equation (14) yields the optimal benchmark for COLA under known factual-counterfactual alignment, as demonstrated in Figure 4 in Section 6.

H EXTENDED NUMERICAL RESULTS

We analyze how different Shapley methods affect FA, corresponding to lines 3–4 in COLA. Comparing RB- p_{Uni} with CF- p_{Uni} in Figures 5–8, CF- p_{Uni} generally performs better, showing the value of using counterfactual information from $\mathbf{r}$ and $\mathbf{r}$ for factual instances. Comparing RB- p_{OT} with CF- p_{OT} , which also differ only in A_{Shap} , CF- p_{OT} consistently outperforms RB- p_{OT} . Thus, across these extended scenarios, *RB-SHAP is not suitable for FA in CE* when the goal is sparse counterfactual refinement. The same trend is visible in the detailed panels in Figures 5–8.

Beyond FA, line 5 is equally important because it uses the joint probability $p(\mathbf{x}, \mathbf{r})$ to construct $\mathbf{q}$, which determines the direction of feature replacement for moving factual instances $\mathbf{x}$ toward the target model outcome using counterfactual references $\mathbf{r}$. In Figures 5–8, CF- p_{OT} consistently outperforms the other methods across the extended experiments. The methods CF- p_{Uni} , CF- p_{Rnd} , and CF- p_{OT} all provide a joint probability $\mathbf{p}$ when exact factual-counterfactual alignment is unknown, but their performance differs substantially. This indicates that having counterfactual examples $\mathbf{r}$, or only their marginal distribution relative to $\mathbf{x}$, is not sufficient; the alignment encoded by $\mathbf{p}$ matters.

OT is particularly useful when alignment information between factual $\mathbf{x}$ and counterfactual $\mathbf{r}$ instances is missing or unreliable. Even when a CE algorithm explicitly matches each factual instance $\mathbf{x}$ to a counterfactual $\mathbf{r}$, that alignment is not necessarily optimal for sparse refinement. Figure 4 in Section 6 supports this observation.

The plan p_{OT} need not be the true data-generating joint distribution of $\mathbf{x}$ and $\mathbf{r}$. Its role is to guide COLA in jointly using factual and counterfactual data $\mathbf{r}$ for both FA and CE. The QDA column in Figure 5 further shows that OT-based methods are stable while several alternatives diverge from the target. At the same time, COLA is *not limited to using OT* as A_{Prob} . As Figure 4 indicates, even a known alignment can have a non-negligible gap to the global optimum, so designing stronger A_{Prob} algorithms remains an important direction.

I EXPERIMENTS REPRODUCIBILITY

The experiments are conducted on a high-performance computing (HPC) cluster, using four nodes in parallel for the four datasets. Each node has two Intel Xeon Processor 2660v3 CPUs (10 core, 2.60GHz) and 128 GB memory. Runs take approximately 5–10 hours per node, depending on dataset size. The experiments can also be reproduced on a laptop, although generally with longer runtime than on the HPC cluster.

For all four datasets, numerical features are standardized, and categorical features use either label encoding or one-hot encoding. In practice, these encoding choices did not materially change COLA’s performance. The train-test split is 7 : 3.

The optimality baseline in Figure 4 is solved with Gurobi 11.0.2 ([gur](#)). Reproducing this baseline requires a Gurobi license, although open-source operations-research libraries such as Google OR-Tools ([goo](#)) can be used as alternatives. The MILP in Appendix G is computationally expensive and is therefore practical only for small-scale settings such as German Credit. For larger datasets, reproducing the optimality baseline requires reducing the number of features.

The model hyperparameters are as follows. Bagging, GP, RBF, RndForest, AdaBoost, GradBoost, and QDA use scikit-learn implementations ([skl](#)) with default hyperparameters. DNN, SVM, RBF, and LR are implemented in PyTorch ([Paszke et al., 2019](#)); the DNN has three layers and the SVM uses a linear kernel. XGBoost ([Chen & Guestrin, 2016](#)) and LightGBM ([Ke et al., 2017](#)) are used through their scikit-learn interfaces with default hyperparameters.

THE USE OF LARGE LANGUAGE MODELS (LLMs)

We used large language models (LLMs) only as general-purpose writing-assistance tools. Their role was limited to grammar, wording, and light copy-editing of author-written text. LLMs did not contribute research ideation, modeling choices, experimental design, datasets, proofs, or results. All algorithms, proofs, datasets, and analyses were created and verified by the authors. Any LLM-suggested phrasing was reviewed and edited before inclusion in the paper. The authors take full responsibility for all content; LLMs are not authors or contributors. This disclosure complies with the ICLR policy on LLM usage.

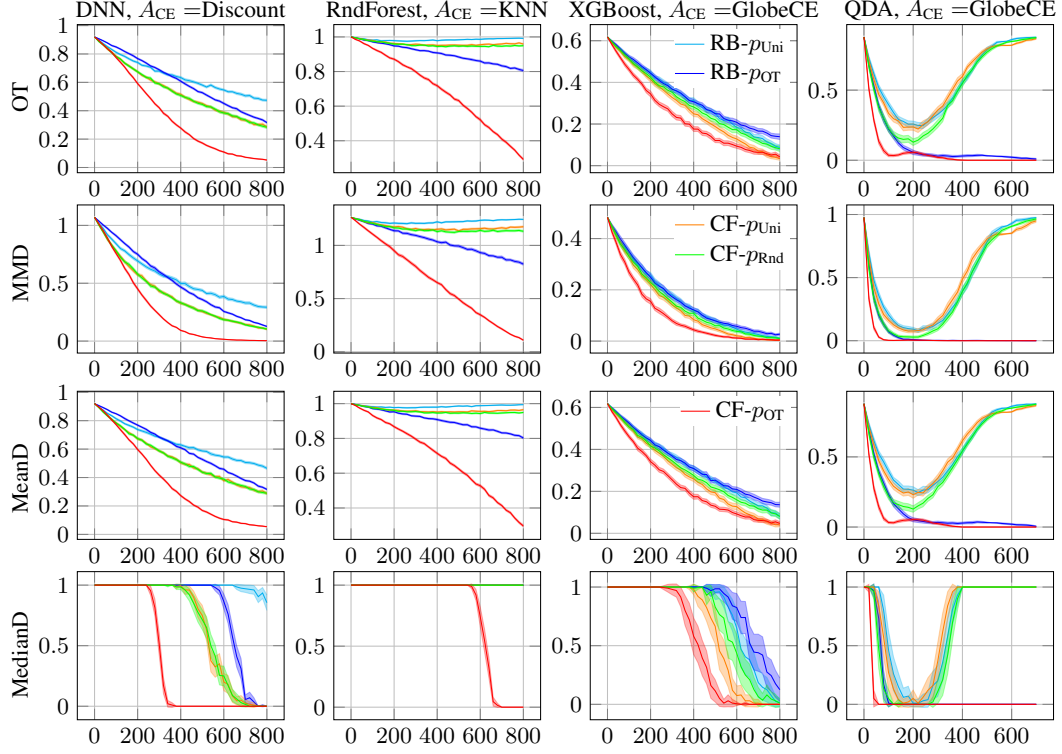


Figure 5: [HELOC] $D(f(\mathbf{z}), \mathbf{y}^*)$ as a function of the action budget C for the evaluated scenarios. Each experiment uses 100 runs; shaded regions show 99.9% confidence intervals. Legends apply to all plots. $A_{\text{Value}}^{\text{avg}}$ is used.

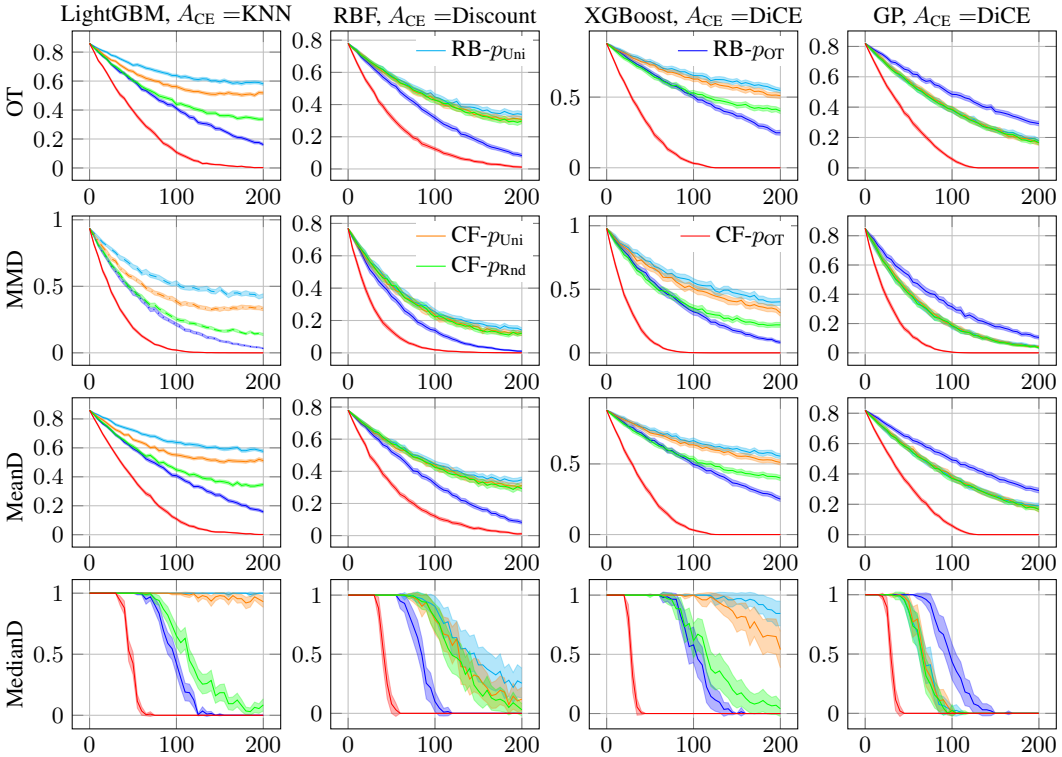


Figure 6: [German Credit] $D(f(\mathbf{z}), \mathbf{y}^*)$ as a function of the action budget C for the evaluated scenarios. Each experiment uses 100 runs; shaded regions show 99.9% confidence intervals. Legends apply to all plots. $A_{\text{Value}}^{\text{avg}}$ is used.

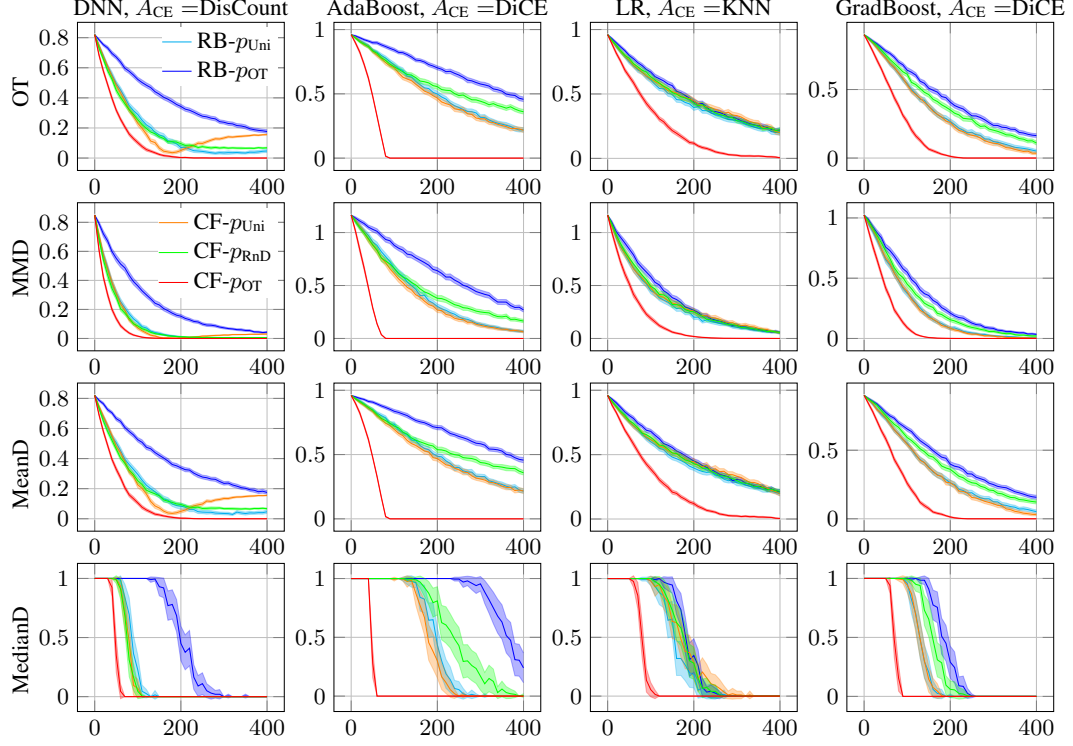


Figure 7: [Hotel Bookings] $D(f(\mathbf{z}), \mathbf{y}^*)$ as a function of the action budget C for the evaluated scenarios. Each experiment uses 100 runs; shaded regions show 99.9% confidence intervals. Legends apply to all plots. $A_{\text{Value}}^{\max}$ is used.

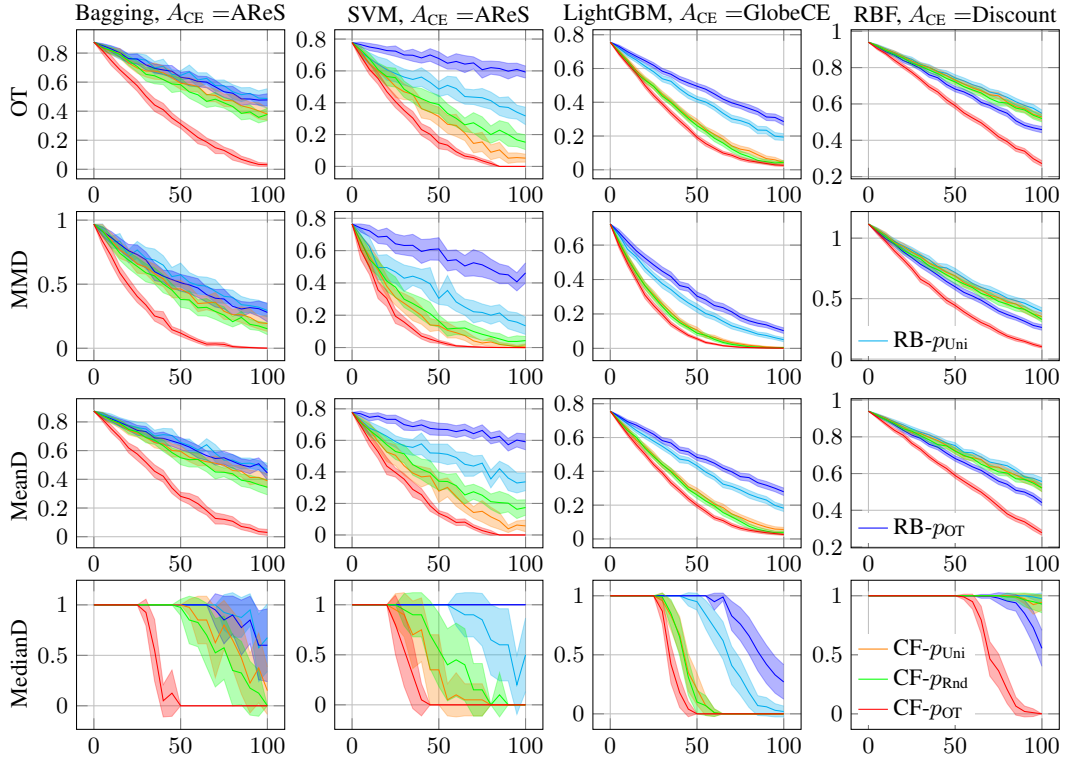


Figure 8: [COMPAS] $D(f(\mathbf{z}), \mathbf{y}^*)$ as a function of the action budget C for the evaluated scenarios. Each experiment uses 100 runs; shaded regions show 99.9% confidence intervals. Legends apply to all plots. $A_{\text{Value}}^{\max}$ is used.